

Hosting a Static Website Theory :

Overview of web hosting with Apache/Nginx. File placement and basic configurations for static sites. Practical (1 hour): Host a simple HTML page on the web server. Task: Customize and host a static webpage.

TASKS

1. Install Apache or Nginx on your Linux server and verify the installation by accessing the default web page via your browser.

Install Apache and Verify Installation

Step 1: Update Packages

`sudo dnf update -y`

```
root@dhcp172-17-172-172:~# sudo dnf update -y
[du@localhost ~]$ sudo dnf update -y
[sudo] password for du:
CentOS Stream 9 - BaseOS                1.9 kB/s | 5.4 kB      00:02
CentOS Stream 9 - AppStream              3.3 kB/s | 5.5 kB      00:01
CentOS Stream 9 - Extras packages        2.8 kB/s | 6.2 kB      00:02
Dependencies resolved.
Nothing to do.
Complete!
[du@localhost ~]$
```

Step 2: Install Apache

`sudo dnf install httpd -y`

```
du@localhost:~  
Installing      : mod_lua-2.4.62-14.el9.x86_64      8/11  
Installing      : centos-logos-httpd-90.9-1.el9.noarch 9/11  
Installing      : mod_http2-2.0.26-6.el9.x86_64     10/11  
Installing      : httpd-2.4.62-14.el9.x86_64        11/11  
Running scriptlet: httpd-2.4.62-14.el9.x86_64        11/11  
Verifying       : apr-1.7.0-12.el9.x86_64           1/11  
Verifying       : apr-util-1.6.1-23.el9.x86_64       2/11  
Verifying       : apr-util-bdb-1.6.1-23.el9.x86_64   3/11  
Verifying       : apr-util-openssl-1.6.1-23.el9.x86_64 4/11  
Verifying       : centos-logos-httpd-90.9-1.el9.noarch 5/11  
Verifying       : httpd-2.4.62-14.el9.x86_64         6/11  
Verifying       : httpd-core-2.4.62-14.el9.x86_64     7/11  
Verifying       : httpd-filesystem-2.4.62-14.el9.noarch 8/11  
Verifying       : httpd-tools-2.4.62-14.el9.x86_64   9/11  
Verifying       : mod_http2-2.0.26-6.el9.x86_64      10/11  
Verifying       : mod_lua-2.4.62-14.el9.x86_64       11/11  
  
Installed:  
apr-1.7.0-12.el9.x86_64      apr-util-1.6.1-23.el9.x86_64  
apr-util-bdb-1.6.1-23.el9.x86_64  
centos-logos-httpd-90.9-1.el9.noarch  
httpd-core-2.4.62-14.el9.x86_64  
httpd-tools-2.4.62-14.el9.x86_64  
mod_lua-2.4.62-14.el9.x86_64  
apr-util-openssl-1.6.1-23.el9.x86_64  
httpd-2.4.62-14.el9.x86_64  
httpd-filesystem-2.4.62-14.el9.noarch  
mod_http2-2.0.26-6.el9.x86_64  
  
Complete!  
[du@localhost ~]$
```

Step 3: Check Apache Version

httpd -v

```
[du@localhost /]$ httpd -v  
Server version: Apache/2.4.62 (CentOS Stream)  
Server built:   May 29 2026 00:00:00  
[du@localhost /]$
```

Step 4: Start Apache

sudo systemctl start httpd

```
Server built:   May 29 2026 00:00:00  
[du@localhost /]$ sudo systemctl start httpd  
[sudo] password for du:
```

Step 5: Enable Apache

sudo systemctl enable httpd

```
[sudo] password for du:  
[du@localhost /]$ sudo systemctl enable httpd  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.  
[du@localhost /]$
```

Step 6:

Verify Service

sudo systemctl status httpd

Step 7:
Open

```
/etc/systemd/system/httpd.service:
[du@localhost ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 10:36:34 IST; 46min ago
     Docs: man:httpd.service(8)
   Main PID: 7836 (httpd)
  Status: "Total requests: 18; Idle/Busy workers 100/0; Requests/sec: 0.00647"
    Tasks: 230 (limit: 4273)
   Memory: 9.0M (peak: 23.0M)
      CPU: 4.060s
   CGroup: /system.slice/httpd.service
           └─7836 /usr/sbin/httpd -DFOREGROUND
             └─7838 /usr/sbin/httpd -DFOREGROUND
               └─7842 /usr/sbin/httpd -DFOREGROUND
                 └─7843 /usr/sbin/httpd -DFOREGROUND
                   └─7845 /usr/sbin/httpd -DFOREGROUND
                     └─8399 /usr/sbin/httpd -DFOREGROUND

Jul 14 10:36:34 localhost.localdomain systemd[1]: Starting The Apache HTTP Serv
Jul 14 10:36:34 localhost.localdomain httpd[7836]: AH00558: httpd: Could not re
Jul 14 10:36:34 localhost.localdomain systemd[1]: Started The Apache HTTP Serve
Jul 14 10:36:34 localhost.localdomain httpd[7836]: Server configured, listening
lines 1-21/21 (END)
```

Firewall

`sudo firewall-cmd --permanent --add-service=http`

```
[du@localhost ~]$ sudo firewall-cmd --permanent --add-service=http
success
```

`sudo firewall-cmd --reload`

```
[du@localhost ~]$ sudo firewall-cmd --reload
success
```

Verify:

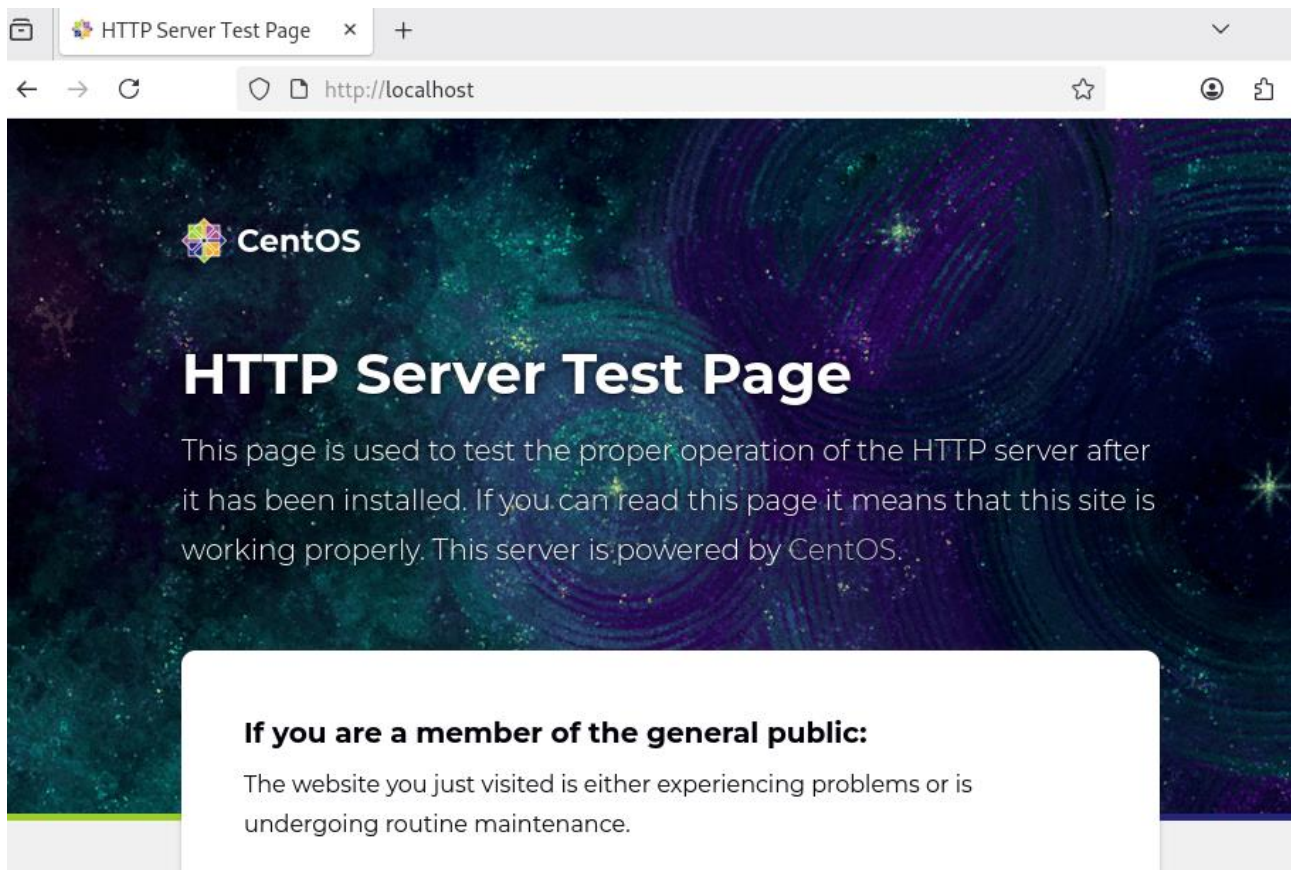
`sudo firewall-cmd --list-services`

```
[du@localhost ~]$ sudo firewall-cmd --list-services
cockpit dhcpv6-client http ssh
```

Step 8: Verify in Browser

Open Firefox and visit:

<http://localhost>



Verify Apache is Running

Command

`sudo systemctl status httpd`

```
du@localhost:~ — sudo systemctl status httpd
[du@localhost ~]$ sudo systemctl status httpd
[sudo] password for du:
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: d>
   Active: active (running) since Tue 2026-07-14 10:36:34 IST; 33min ago
     Docs: man:httpd.service(8)
  Main PID: 7836 (httpd)
    Status: "Total requests: 18; Idle/Busy workers 100/0;Requests/sec: 0.00908>
       Tasks: 230 (limit: 4273)
      Memory: 9.0M (peak: 23.0M)
         CPU: 2.991s
    CGroup: /system.slice/httpd.service
            └─7836 /usr/sbin/httpd -DFOREGROUND
              └─7838 /usr/sbin/httpd -DFOREGROUND
                └─7842 /usr/sbin/httpd -DFOREGROUND
                  └─7843 /usr/sbin/httpd -DFOREGROUND
                    └─7845 /usr/sbin/httpd -DFOREGROUND
                      └─8399 /usr/sbin/httpd -DFOREGROUND

Jul 14 10:36:34 localhost.localdomain systemd[1]: Starting The Apache HTTP Serv>
Jul 14 10:36:34 localhost.localdomain httpd[7836]: AH00558: httpd: Could not re>
Jul 14 10:36:34 localhost.localdomain systemd[1]: Started The Apache HTTP Serve>
Jul 14 10:36:34 localhost.localdomain httpd[7836]: Server configured, listening>
lines 1-21/21 (END)
```

2. Create a folder named myflipkart in your web server's root directory and place an index.html file inside it that displays 'Welcome to MyFlipkart Deals!'.

Create "myflipkart" Website

Step 1: Go to Apache Root

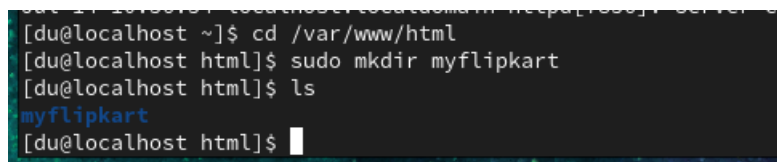
```
cd /var/www/html
```

Step 2: Create folder:

```
sudo mkdir myflipkart
```

Check:

```
ls
```

A terminal window screenshot showing the following commands and output: [du@localhost ~]\$ cd /var/www/html, [du@localhost html]\$ sudo mkdir myflipkart, [du@localhost html]\$ ls, myflipkart, [du@localhost html]\$.

[du@localhost ~]\$ cd /var/www/html
[du@localhost html]\$ sudo mkdir myflipkart
[du@localhost html]\$ ls
myflipkart
[du@localhost html]\$

Step 3: Create HTML File

```
sudo vi /var/www/html/myflipkart/index.html
```

Press i

Paste:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>My Flipkart</title>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to MyFlipkart Deals!</h1>
```

```
<p>This website is hosted using Apache Web Server.</p>
```

```
</body>
```

```
</html>
```

Save:

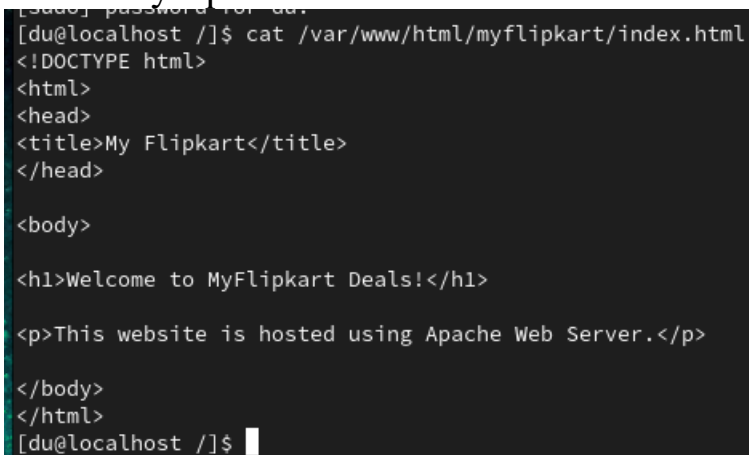
Esc

:wq

Enter

Verify:

cat /var/www/html/myflipkart/index.html

A terminal window with a dark background and light green text. The prompt is [du@localhost ~]\$. The command cat /var/www/html/myflipkart/index.html has been executed, displaying the following HTML content:

```
<!DOCTYPE html>
<html>
<head>
<title>My Flipkart</title>
</head>

<body>

<h1>Welcome to MyFlipkart Deals!</h1>

<p>This website is hosted using Apache Web Server.</p>

</body>
</html>
[du@localhost ~]$
```

3. Update your web server configuration so that visiting <http://localhost/myflipkart> shows your custom index.html page instead of the default server page.
Hint: For Apache, use the Alias directive; for Nginx, use a new location block.

Configure Apache

Open Apache configuration:

```
sudo vi /etc/httpd/conf/httpd.conf
```

Go to the bottom:

Shift + G

Press i**Add:**

```
Alias /myflipkart "/var/www/html/myflipkart"
```

```
<Directory "/var/www/html/myflipkart">
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>
```

Save:

Esc

:wq

Restart Apache:

sudo systemctl restart httpd

Verify:

sudo systemctl status httpd

```
[du@localhost ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 11:41:15 IST; 8s ago
     Docs: man:httpd.service(8)
   Main PID: 10495 (httpd)
   Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 4273)
   Memory: 14.6M (peak: 14.8M)
      CPU: 133ms
   CGroup: /system.slice/httpd.service
           └─10495 /usr/sbin/httpd -DFOREGROUND
             └─10496 /usr/sbin/httpd -DFOREGROUND
               └─10500 /usr/sbin/httpd -DFOREGROUND
                 └─10501 /usr/sbin/httpd -DFOREGROUND
                   └─10503 /usr/sbin/httpd -DFOREGROUND

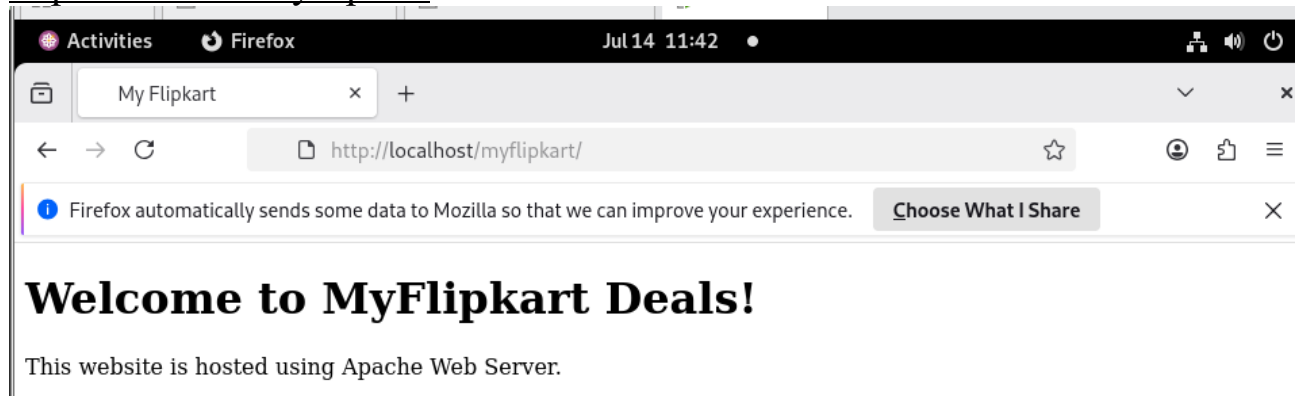
Jul 14 11:41:15 localhost.localdomain systemd[1]: Starting The Apache HTTP Serv
Jul 14 11:41:15 localhost.localdomain httpd[10495]: AH00558: httpd: Could not r
Jul 14 11:41:15 localhost.localdomain systemd[1]: Started The Apache HTTP Serve
Jul 14 11:41:15 localhost.localdomain httpd[10495]: Server configured, listenin
lines 1-20/20 (END)
```

Open

Browser

Visit:

<http://localhost/myflipkart>



4. Add a favicon to your static site by placing a favicon.ico file in the correct location and updating your HTML to reference it.
Constraint: Use any small image and rename it to favicon.ico for this exercise.

Step 1– Use Firefox

1. Open **Firefox**.
2. Search for a small icon (for example, a Flipkart logo).
3. Save the image to the **Downloads** folder.

Check it:

```
cd ~/Downloads
```

```
ls
```

If you see:

```
flipkart.png
```

Copy the Image to Your Website Folder:

```
sudo cp ~/Downloads/flipkart.jpg /var/www/html/myflipkart/favicon.ico
```

Step 2: Verify

Run:

```
ls /var/www/html/myflipkart
```

```
[du@localhost Downloads]$ ls /var/www/html/myflipkart  
favicon.ico  index.html
```

Step 3: Edit index.html

Open the HTML file:

```
sudo vi /var/www/html/myflipkart/index.html
```

Step 4: Press i

This enters Insert Mode.

Step 5: Add the Favicon Line

Inside the `<head>` section, change it to:

```
<head>
```

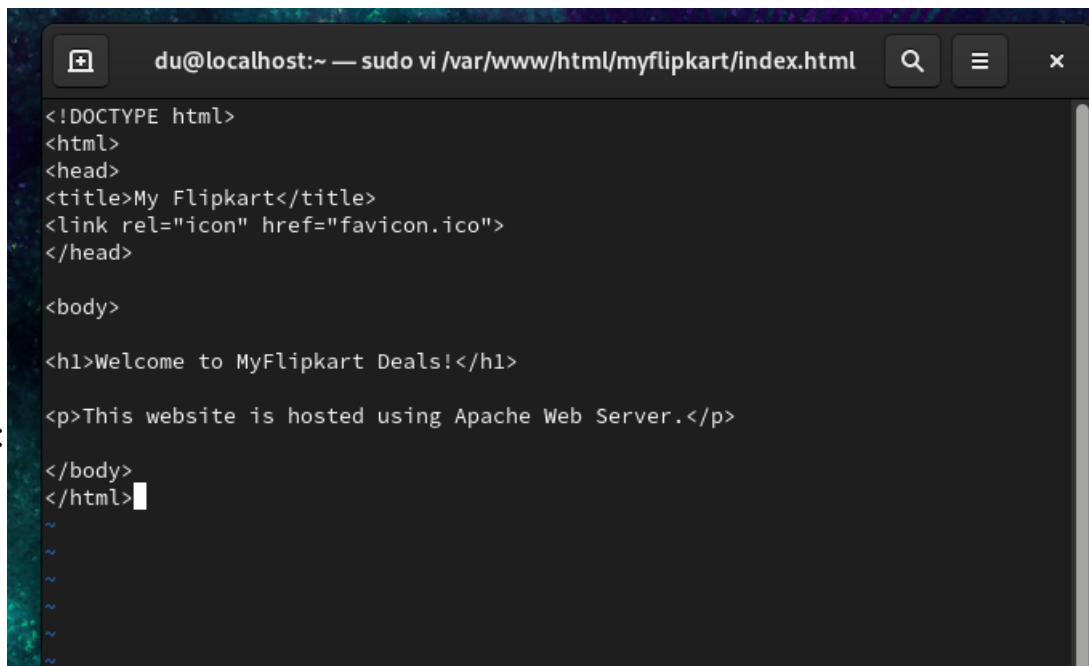
```
<title>My Flipkart</title>
```

```
<link rel="icon" href="favicon.ico">
```

```
</head>
```


Step 6: Save the File

Press:
ESC



```
du@localhost:~ — sudo vi /var/www/html/myflipkart/index.html
<!DOCTYPE html>
<html>
<head>
<title>My Flipkart</title>
<link rel="icon" href="favicon.ico">
</head>

<body>

<h1>Welcome to MyFlipkart Deals!</h1>

<p>This website is hosted using Apache Web Server.</p>

</body>
</html>
```

Type:- :wq

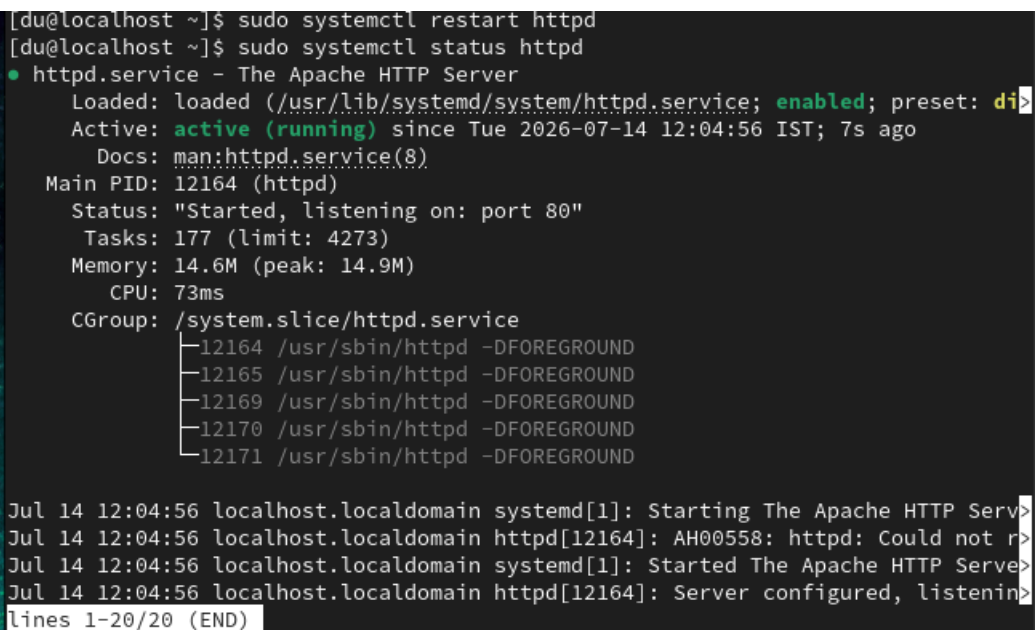
Press Enter

Step 7: Restart Apache

sudo systemctl restart httpd

Check Status:

sudo systemctl status httpd



```
[du@localhost ~]$ sudo systemctl restart httpd
[du@localhost ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 12:04:56 IST; 7s ago
     Docs: man:httpd.service(8)
  Main PID: 12164 (httpd)
    Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 4273)
   Memory: 14.6M (peak: 14.9M)
      CPU: 73ms
   CGroup: /system.slice/httpd.service
           └─12164 /usr/sbin/httpd -DFOREGROUND
             └─12165 /usr/sbin/httpd -DFOREGROUND
               └─12169 /usr/sbin/httpd -DFOREGROUND
                 └─12170 /usr/sbin/httpd -DFOREGROUND
                   └─12171 /usr/sbin/httpd -DFOREGROUND

Jul 14 12:04:56 localhost.localdomain systemd[1]: Starting The Apache HTTP Serv>
Jul 14 12:04:56 localhost.localdomain httpd[12164]: AH00558: httpd: Could not r>
Jul 14 12:04:56 localhost.localdomain systemd[1]: Started The Apache HTTP Serve>
Jul 14 12:04:56 localhost.localdomain httpd[12164]: Server configured, listenin>
lines 1-20/20 (END)
```

Step 8: Open the Website

Open **Firefox**.

Visit:

<http://localhost/myflipkart>

